

Rayan Mohammed

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PROFILE

CS final-year student (VIT-AP, expected 2026) who builds and ships full-stack features end-to-end. Designed a hybrid DQN + NSGA-II scheduling algorithm that beat six optimization baselines by 38–44% ($p < 0.001$), and ships backend REST systems and automation pipelines (Docker, n8n). Strong in data structures, algorithms, distributed systems, and problem-solving.

SKILLS

Programming: Python, Java, JavaScript, SQL

Backend: Django, REST API Development, MySQL

Tools: Git, GitHub, Linux, Docker, Docker Compose, n8n

Cloud: AWS (EC2, S3, IAM, VPC, Auto Scaling, ELB)

Data & Analytics: Pandas, NumPy, PyTorch, Scikit-learn

Core Concepts: Data Structures, Algorithms, OOP, DBMS, Distributed Systems

EXPERIENCE

CVS Info Solutions, Infopark

July 2024

Software Development Intern

- Resolved authentication, data-flow, and user-access defects by debugging REST API endpoints and backend modules across the application stack
- Secured user and admin workflows by implementing and analyzing role-based access control (RBAC)
- Improved database query performance by investigating API failures through log analysis
- Diagnosed and fixed functional and UI defects by collaborating with developers to reproduce issues, sharpening problem-solving and analytical thinking
- Streamlined debugging and testing cycles by documenting recurring issues and solutions

MicroIT – Student Tech Initiative

December 2023 – January 2025

Co-Founder

- Shipped core features for a live B2B travel platform (VayoAura) serving real customers, owning backend development end-to-end in a startup-pace environment
- Led API design and system architecture for web and mobile solutions
- Translated client requirements into backend workflows, taking initiative from spec to deployment
- Drove production readiness by owning debugging, testing, and iteration cycles

AIESEC in Amaravati

August 2024 – May 2025

Business Development Associate

- Strengthened communication with decision-makers by pitching tailored value propositions and handling objections, qualifying leads by fit and intent
- Booked 5–6 discovery calls and partner meetings per month for senior team members
- Engaged 40–60 B2B prospects per week by running outbound prospecting across email, LinkedIn, and cold calls
- Sustained consistent engagement across multi-touch sequences by maintaining a structured lead pipeline with disciplined follow-ups

PROJECTS

Task Scheduling Optimization in Distributed Systems Python, PyTorch, NumPy, Alibaba Cluster Trace Dataset

- Outperformed GA, PSO, GWO, EHO, HBO, and RL baselines by 38–44% ($p < 0.001$, Wilcoxon rank-sum, 30 runs) by designing a hybrid DQN + NSGA-II + Honey Badger Optimization algorithm for multi-objective task scheduling on real Alibaba cluster traces
- Achieved HEFT-optimal makespan on a 450-task, 10-processor concurrent workload by building a Deep Q-Network manager that adaptively selects among 7 evolutionary operators using experience replay and ϵ -greedy exploration
- Validated multi-objective convergence across ZDT (1–6) and DTLZ (1–7) benchmark suites, producing well-spread Pareto fronts with up to 50 non-dominated solutions per configuration, applying advanced data structures & algorithms
- Engineered an end-to-end DAG-based scheduling simulator with HEFT priority ranking, SLA deadline enforcement, and joint optimization of makespan, energy, wait time, and processor utilization

Clara AI Automation Pipeline

Docker, Docker Compose, n8n, JSON, Workflow Automation

- Automated processing of AI call transcripts into structured agent-configuration artifacts by building a containerized automation pipeline with Docker Compose and n8n
- Designed version-controlled workflows for automated data extraction and JSON-based account-memo generation
- Built modular orchestration workflows for transcript ingestion, transformation, and artifact generation with local containerized deployment
- Improved reproducibility and debugging through isolated service deployment and structured workflow automation

VSync – Smart College Companion Platform

Flutter, Dart, REST API

- Built a modular full-stack academic-management platform by developing Flutter front-end and REST API integrations
- Enabled accurate GPA computation by designing structured data models for academic records
- Powered dynamic data retrieval and state management by integrating backend services
- Implemented reusable UI components using clean-architecture principles

Gesture-Controlled Drone

Raspberry Pi, OpenCV, Python

- Developed a real-time computer vision system to map hand gestures to drone flight controls
- Implemented gesture recognition pipelines using OpenCV for accurate motion detection
- Optimized image processing for low-latency response
- Designed for surveillance, search operations, and hands-free filming use cases

EDUCATION

Vellore Institute of Technology Andhra Pradesh (VIT-AP)

2022 – 2026 (Expected May 2026)

B.Tech in Computer Science Engineering

CERTIFICATIONS

- [AWS Academy Cloud Architecting \(2024\)](#)
- [AWS Academy Cloud Foundations \(2024\)](#)
- [CouchDB Distributed Database System \(2024\)](#)